

Spring Boot @Scheduled Quick Notes

Enable Scheduling

```
@EnableScheduling
```

Syntax

```
@Scheduled(cron = "second minute hour day-of-month month day-of-week")
```

Cron Format (6 fields):

```

|----- Second (0-59)
| |----- Minute (0-59)
| | |----- Hour (0-23)
| | | |----- Day of Month (1-31)
| | | | |----- Month (1-12)
| | | | | |----- Day of Week (0-7 or SUN-SAT)
* * * * *

```

Remember:

Second → Minute → Hour → Day → Month → Week

Common Symbols

Symbol	Meaning	Example
*	Every value	* = every minute/hour/day
*/n	Every n units	*/5 = every 5 minutes
,	Multiple values	1,15,30
-	Range	1-5

Most Used Cron Expressions

Every 5 seconds

```
@Scheduled(cron = "*/5 * * * * *")
```

Every minute

```
@Scheduled(cron = "0 */1 * * * *")
```

Every 5 minutes

```
@Scheduled(cron = "0 */5 * * * *")
```

Every 15 minutes

```
@Scheduled(cron = "0 */15 * * * *")
```

Every hour

```
@Scheduled(cron = "0 0 * * * *")
```

Every day at 2:30 AM

```
@Scheduled(cron = "0 30 2 * * *")
```

Every Monday at 9:00 AM

```
@Scheduled(cron = "0 0 9 * * MON")
```

fixedRate vs fixedDelay

Every 60 seconds (start-to-start)

```
@Scheduled(fixedRate = 60000)
```

- Runs every 60 seconds.
- Doesn't wait for the previous execution to finish.

Every 60 seconds (finish-to-start)

```
@Scheduled(fixedDelay = 60000)
```

- Waits until the previous execution finishes.
 - Then waits 60 seconds before running again.
-

Easy Memory Trick

```
Cron = Calendar Time  
fixedRate = Every X milliseconds  
fixedDelay = Wait X milliseconds after completion
```

Mnemonic:

- **Cron** → "When?"
 - **fixedRate** → "Every X ms"
 - **fixedDelay** → "After finishing, wait X ms"
-

Example Timeline

```
Cron (Every minute)  
10:00:00  
10:01:00  
10:02:00  
  
fixedRate(60000)  
Start → 60s → Start → 60s → Start  
  
fixedDelay(60000)  
Start → Finish → Wait 60s → Start
```